

EE3621: Digital Signal Processing

Lecture 22: FIR Filter Design via Windowing (Rectangular, Bartlett, Hann) (III B.Tech EEE)

September 9, 2026

1. LEARNING OBJECTIVES

By the end of this lecture, students will be able to:

1. **Derive** ideal impulse responses $h_d[n]$ for Lowpass, Highpass, Bandpass, and Bandstop filters.
 2. **Explain** the origins and mathematical properties of the Gibbs phenomenon.
 3. **Design** linear-phase FIR filters using Rectangular, Bartlett, and Hann windows to meet transition bandwidth and attenuation specs.
 4. **Evaluate** the frequency spectrum of window functions.
-

2. MATHEMATICAL FOUNDATIONS

2.1 Ideal Filter Impulse Responses ($\tau = \frac{N-1}{2}$)

- **Lowpass** (ω_c):

$$h_{d,LP}[n] = \frac{\sin(\omega_c(n - \tau))}{\pi(n - \tau)}$$

- **Highpass** (ω_c):

$$h_{d,HP}[n] = \delta[n - \tau] - \frac{\sin(\omega_c(n - \tau))}{\pi(n - \tau)}$$

- **Bandpass** (ω_{c1}, ω_{c2}):

$$h_{d,BP}[n] = \frac{\sin(\omega_{c2}(n - \tau))}{\pi(n - \tau)} - \frac{\sin(\omega_{c1}(n - \tau))}{\pi(n - \tau)}$$

2.2 Window Functions Formulation ($0 \leq n \leq N - 1$)

1. Rectangular Window:

$$w[n] = 1, \quad \Delta\omega = \frac{4\pi}{N}, \quad \text{Peak Sidelobe: } -13 \text{ dB}, \quad A_s = 21 \text{ dB}$$

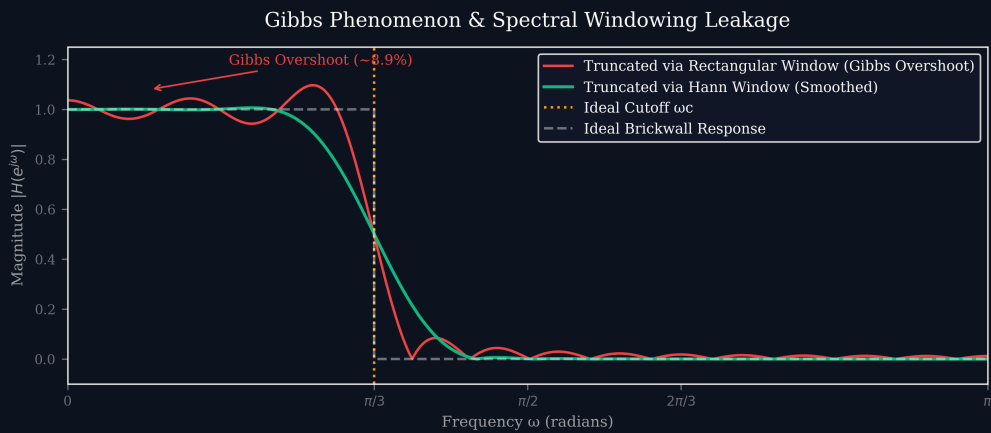
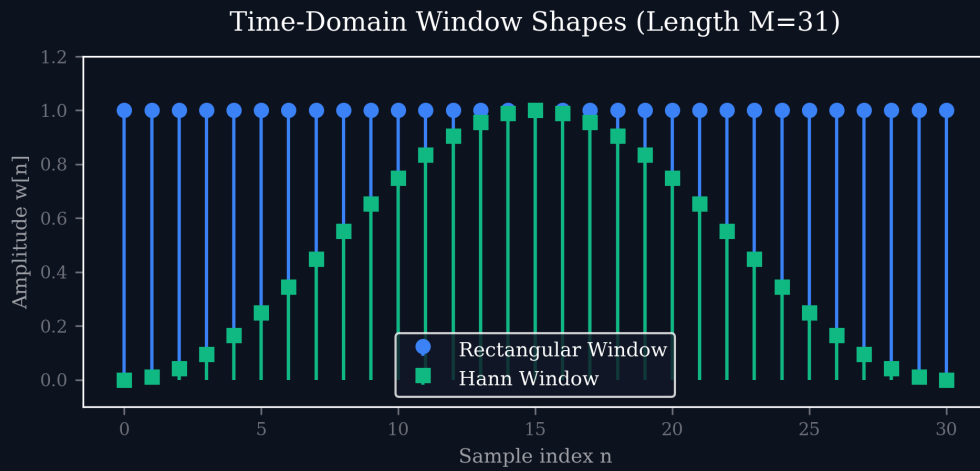
1. Bartlett (Triangular) Window:

$$w[n] = 1 - \frac{2|n - \tau|}{N - 1}, \quad \Delta\omega = \frac{8\pi}{N}, \quad \text{Peak Sidelobe: } -25 \text{ dB}, \quad A_s = 25 \text{ dB}$$

1. Hann (Hanning) Window:

$$w[n] = 0.5 - 0.5 \cos\left(\frac{2\pi n}{N - 1}\right), \quad \Delta\omega = \frac{8\pi}{N}, \quad \text{Peak Sidelobe: } -31 \text{ dB}, \quad A_s = 44 \text{ dB}$$

Pictorial Representations



3. WORKED NUMERICAL EXAMPLES

Example 22.1: FIR Lowpass Filter Design using Hann Window

Problem: Design a causal FIR lowpass filter with cutoff $\omega_c = \frac{\pi}{4}$ rad/sample and length $N = 7$ using a Hann window.

Solution:

- Length $N = 7 \implies \tau = \frac{7-1}{2} = 3$.

- Ideal impulse response:

$$h_d[n] = \frac{\sin(\frac{\pi}{4}(n-3))}{\pi(n-3)}$$

$$\begin{aligned} * h_d[3] &= \frac{\omega_c}{\pi} = \frac{\pi/4}{\pi} = 0.2500 & * h_d[2] &= h_d[4] = \frac{\sin(-\pi/4)}{-\pi} = \frac{\sqrt{2}/2}{\pi} = \frac{0.7071}{\pi} = 0.2251 & * h_d[1] &= \\ h_d[5] &= \frac{\sin(-2\pi/4)}{-2\pi} = \frac{1}{2\pi} = 0.1592 & * h_d[0] &= h_d[6] = \frac{\sin(-3\pi/4)}{-3\pi} = \frac{0.7071}{3\pi} = 0.0750 \end{aligned}$$

- **Hann Window Values** ($w[n] = 0.5 - 0.5 \cos(\frac{2\pi n}{6})$):

$$\begin{aligned} * w[3] &= 0.5 - 0.5 \cos(\pi) = 1.0000 & * w[2] &= w[4] = 0.5 - 0.5 \cos(2\pi/3) = 0.5 - 0.5(-0.5) = 0.7500 \\ * w[1] &= w[5] = 0.5 - 0.5 \cos(\pi/3) = 0.5 - 0.5(0.5) = 0.2500 & * w[0] &= w[6] = 0.5 - 0.5 \cos(0) = 0.0000 \end{aligned}$$

- **Final FIR Filter Coefficients** ($h[n] = h_d[n] \cdot w[n]$):

$$\begin{aligned} * h[0] &= h[6] = 0.0750 \times 0.0000 = 0.0000 & * h[1] &= h[5] = 0.1592 \times 0.2500 = 0.0398 & * h[2] &= \\ h[4] &= 0.2251 \times 0.7500 = 0.1688 & * h[3] &= 0.2500 \times 1.0000 = 0.2500 \end{aligned}$$

$$h[n] = \{0.0000, 0.0398, 0.1688, 0.2500, 0.1688, 0.0398, 0.0000\}$$

4. UNIVERSITY EXAMINATION QUESTIONS & MARKING RUBRIC

Question 1 (15 Marks)

(a) What is the Gibbs phenomenon? Explain how window functions reduce Gibbs ringing at the expense of transition bandwidth. *(6 Marks)* (b) Design a 9-tap FIR Highpass filter with cutoff $\omega_c = \frac{\pi}{3}$ using a Bartlett window. *(9 Marks)*

Model Answer & Step-by-Step Marking Rubric:

- **Part (a):**

* Explanation of 8.95* Tapering effect of windows: Smooth boundary transitions suppress high-frequency sidelobes but widen mainlobe transition band *(3 Marks)*

- **Part (b):**

* $N = 9, \tau = 4$. * $h_d[n] = \delta[n-4] - \frac{\sin(\pi(n-4)/3)}{\pi(n-4)}$ *(3 Marks)* * Bartlett window: $w[n] = 1 - \frac{|n-4|}{4}$ for $n = 0, \dots, 8$ *(2 Marks)* * Evaluation of all 9 coefficients $h[n] = h_d[n]w[n]$ with symmetry *(4 Marks)*

5. PYTHON VERIFICATION SCRIPT

```
import numpy as np
```

```
N = 7
```

```
tau = 3
```

```
wc = np.pi / 4
```

```
n = np.arange(N)
```

```
\title{\textbf{EE3621: Digital Signal Processing}} \Large Ideal LPF \large (III B.Tech
```

```
hd = np.zeros(N)
```

```
hd[n != tau] = np.sin(wc * (n[n != tau] - tau)) / (np.pi * (n[n != tau] - tau))
```

```
hd[tau] = wc / np.pi
```

```
\title{\textbf{EE3621: Digital Signal Processing}} \Large Hann window \large (III B.Te
```

```
w = 0.5 - 0.5 * np.cos(2 * np.pi * n / (N - 1))
```

```
h = hd * w
```

```
print("Designed FIR Filter Coefficients h[n]:")
```

```
print(np.round(h, 4))
```